Course: BTech **Semester:** 3

Prerequisite: Basic knowledge of Problem Solving,

Rationale : This course introduces Java programming and object-oriented concepts to build a strong foundation for developing reliable, platform-independent software applications and for pursuing advanced studies in software development.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
3	-	-	-	3	20	20	-	60	-	100

SEE - Semester End Examination, **T** - Theory, **P** - Practical

Course Content

W - Weightage (%) , **T** - Teaching hours

Sr.	Topics	W	T
1	**Overview of OOP and Java:**Programming Paradigms, Procedural vs Object-Oriented Programming, OOP Features, History of Java, Feature of Java, JVM Architecture, JRE, JDK, Java Program Structure, Compilation and Execution Process, Java Environment Setup, JDK Installation, PATH and CLASSPATH, Introduction to IDEs.	10	4
2	**Java Language Basics:**Variables, Scope of Variables, Primitive Data Types, Reference Data Types, Type Casting, Wrapper Classes, Autoboxing and Unboxing, Arithmetic Operators, Relational Operators, Logical Operators, Assignment Operators, Unary Operators, Bitwise Operators, Operator Precedence, if Statement, if-else Statement, Nested if, switch Statement, for Loop, while Loop, do-while Loop, for - Each Loop, break Statement, continue Statement.	15	7
3	**Arrays and String**Arrays, One-Dimensional Arrays, Multi-Dimensional Arrays, Array Operations, String Class, StringBuffer, StringBuilder, String Comparison, String Manipulation Methods	12	6
4	**Object-Oriented Programming (OOP)**Classes and Objects, Constructors, Access Specifiers, Encapsulation, Inheritance, Types of Inheritance, super Keyword, Polymorphism, Method Overloading(Compile time Polymorphism), Method Overriding (Runtime Polymorphism), Abstraction, Abstract Classes, Interfaces, Functional Interfaces, this Keyword, final Keyword, Object Class and its Methods.	30	12
5	**Exception Handling**Errors and Exceptions, Checked Exceptions, Unchecked Exceptions, try-catch Block, Multiple catch Blocks, finally Block, throw Keyword, throws Keyword, Custom Exceptions, Exception Handling, Best Practices	15	6
6	**Collection Frame work in Java**Introduction to Collections Framework, List Interface, ArrayList, LinkedList, Set Interface, HashSet, TreeSet, Map Interface, HashMap, TreeMap, Iterators, Generics	10	5
7	**Advanced Foundations and Java Frameworks**Process Vs Thread, Thread Creation, Thread Life Cycle, Multithreading Concepts, Synchronization, Inter-thread Communication, File Handling, File Class, Byte Streams, Character Streams, Buffered Streams, Serialization, Deserialization, Lambda Expressions, Stream API, Optional Class.	8	5
	Total	100	45

Reference Books	
1.	**Java: The Complete Reference** By Herbert Schildt | Tata McGraw Hill | 7TH
2.	**Object Oriented Programming Through Java — P. Radha Krishna Publisher: Universities Press (India) Pvt. Ltd. (TextBook)**
3.	**Thinking in Java — Bruce Eckel Publisher: Prentice Hall (Pearson)**
4.	**Core Java, Volume I & Volume II — Cay S. Horstmann & Gary Cornell Publisher: Prentice Hall / Pearson**
5.	**JAVA8 Core Java Black Book — R. Nageswara Rao Publisher: Black Book Series (Dreamtech Press) (TextBook)**

Course Outcome

After Learning the Course the students shall be able to:

1. Develop and execute basic Java programs by following standard program structure and compilation steps.
2. Apply arithmetic, relational, logical, assignment, unary, and bitwise operators while considering operator precedence.
3. Create and manipulate one-dimensional and multi-dimensional arrays and perform common array operations in Java.
4. Analyze and handle runtime errors using Java exception handling mechanisms to develop robust and reliable applications.
5. Design and use custom exceptions and apply exception-handling best practices in Java programs.
6. Develop Java applications using advanced features such as multithreading, file handling, modern Java APIs, and demonstrate awareness of Java frameworks.